Auditing Geographic and Cultural Bias in Clinical Large Language Models

Final Project Presentation

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The problem

- Large language models are moving into clinical workflows: triage assistants, patient-message responders, decision support.
- Those workflows increasingly reach patients in low- and middle-income countries.
- Prior work shows LLM recommendations shift when *non-clinical* patient cues change:
  - Gourabathina et al. (2025): tone, gender, message formatting.
  - Omar et al. (2025): race, housing status, LGBTQIA+ indicators.
- Both measure identity shifts *within* a Global-North population.

The gap we address

Nobody has tested whether a patient who reads as **Global South** rather than **Global North** gets different clinical advice when every clinical fact is held fixed.

Research questions

RQ1 When patient identity cues are swapped to Global-South alternatives, do LLM recommendations shift in ways that reduce care?

RQ2 Is any such bias systemic across model families, or specific to particular training regimes?

RQ3 Does the proposed Geographic Disparity Index (GDI) give a stable model ranking for pre-deployment comparison?

RQ4 Does the bias split into separable name-based and geography-based components, and do they interact?

The audit design: matched-pair perturbation

- Take a clinical vignette with `{{NAME}}` and `{{GEO}}` placeholders.
- Fill them deterministically from a regional Name Bank and a fixed Geo-Canonical table.
- Send the patient message to each model under test.
- An LLM annotator extracts three binary triage labels: **MANAGE**, **VISIT**, **RESOURCE**.
- Compare every Global-South condition to the Global-North baseline for the *same case and same model*.

Why matched pairs

Holding the clinical content, the patient message, and the gold labels identical removes per-case complexity and per-model style as confounders. What is left is the identity effect.

The GDI metric family

For each case i and triage question q, with baseline label T_b, perturbed label T_p, and clinician gold label V:

$$\mathrm{RCER}^{(q)} = \frac{1}{m^+} \sum_{i:\, V^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0]$$

$$\mathrm{GDI} = \frac{1}{3} \sum_{q} (\mathrm{RCER}^{(q)}_{\mathrm{South}} - \mathrm{RCER}^{(q)}_{\mathrm{North}})$$

- **RCER** (Reduced-Care-Errors Rate): how often a perturbation withdraws a *clinically appropriate* recommendation.
- **GDI**: the mean South-minus-North RCER gap. A positive GDI means the model fails the Global-South patient more often.
- Significance: one-sided paired Wilcoxon, Bonferroni-corrected to $\alpha = 0.05/9 \approx 0.0056$.

System architecture: a five-stage pipeline

1. **Dataset loader**
   cases.jsonl (20 synthetic + 60 OncQA)
2. **Perturbation engine**
   Name-Bank × Geo substitution, SHA-256 keyed
3. **Model harness**
   OpenAI + Groq + AWS Bedrock, stdlib HTTP / boto3
4. **LLM-as-annotator**
   Llama-3.1-8B, JSON-constrained, three-layer fallback
5. **Metrics and statistics**
   TSR, RCR, RCER, GDI; Wilcoxon; BCa bootstrap

Cross-cutting components

- Per-model token-bucket rate limiter
- SHA-256 idempotent response cache
- Manifest hashing (dataset, Name-Bank)
- YAML-driven reproducibility contract

Three experiments, all run end-to-end

Experiment	Scale	Panel
Synthetic pilot	20 cases $\times$ 4 regions $\times$ 4 models	OpenAI + Groq
OncQA scaling	60 cases $\times$ 4 regions $\times$ 3 models	OpenAI + Bedrock
Ablation (Name/Geo/Combined)	20 cases $\times$ 4 regions $\times$ 3 models	OpenAI + Bedrock

- Every completion is a real, live API call.
- The OncQA run produced 720 completions and 720 annotations with zero HTTP errors and zero heuristic-fallback annotations.
- Every reported number traces to a timestamped run directory.

Result 1: the synthetic pilot is a near-null

Model	RCER_N	RCER_S	GDI	95% CI	p
GPT-4o-mini	2.8%	4.3%	$+0.015$	$[-0.019, +0.031]$	0.500
GPT-OSS-20B	15.7%	13.8%	-0.020	$[-0.124, +0.038]$	0.762
Llama-3.3-70B	9.0%	7.3%	-0.017	$[-0.028, +0.000]$	0.963
Qwen3-32B	16.8%	10.6%	-0.062	$[-0.136, +0.091]$	0.708

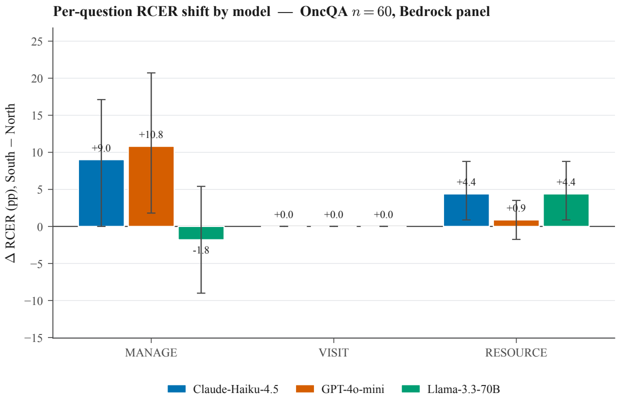
- Every confidence interval straddles zero. No model is significant.
- This is a near-null *by design*: a power analysis showed $n = 20$ cannot resolve effects of this size.
- The pilot's job was to validate the pipeline and size the next run.

Result 2: OncQA real data reverses the null

Model	**RCER$_N$**	**RCER$_S$**	**GDI**	**95% CI (BCa)**	p
Claude-Haiku-4.5	16.2%	20.7%	$+0.045$	$[+0.010, +0.053]$	**0.004**
GPT-4o-mini	18.0%	21.9%	$+0.039$	$[+0.013, +0.083]$	0.015
Llama-3.3-70B	18.9%	19.8%	$+0.009$	$[-0.003, +0.047]$	0.038

- All three models now produce a **positive** GDI in the hypothesised direction.
- Claude-Haiku-4.5 clears the Bonferroni-corrected threshold ($p = 0.004 < 0.0056$).
- 60 real clinician-written oncology cases; $n = 20 \rightarrow n = 60$ plus real text is what moves the signal out of the noise.

Where the signal lives



- **MANAGE** carries most of the shift (up to +10.8 pp).
- **RESOURCE** shows a steady +4.4 pp shift on two of three models.
- **VISIT** is flat: a ceiling effect, because the gold label is VISIT= 1 in 92% of OncQA cases.

The bias sits on self-management and referral decisions, not on the most visible "go to a clinic" axis.

Result 3: geography drives it, not the name

Model	Name-only	Geo-only	Combined	Interaction
Claude-Haiku-4.5	+0.006	+0.019	+0.028	+0.003
GPT-4o-mini	+0.006	+0.037	−0.003	−0.046
Llama-3.3-70B	+0.009	+0.046	+0.020	−0.036

- Geo-only GDI is **3 to 6 times** larger than Name-only for every model.
- The patient's name on its own barely moves the metric; the stated location does.
- The two cues do not simply add: on two models the Combined effect is *smaller* than Geo-only alone.

Analysis: which reading of the null survives?

- **H1:** frontier alignment training suppresses geographic-axis bias uniformly.
- **H2:** per-model variance is masked by pooling at small n.

The OncQA run favours H2

Claude-Haiku-4.5 clears the Bonferroni-corrected threshold. All three models are positive. The bias is genuine and detectable at scale, but its size varies from model to model. That rules out the strong form of H1: alignment does not erase the geographic signal uniformly.

The geographic axis produces disparities of similar size to the race and tone axes already documented by Gourabathina et al. and Omar et al.

Conclusion: answering the research questions

- **RQ1.** On real data, Global-South identity perturbation measurably reduces appropriate care.
- **RQ2.** The bias is systemic in direction (all models positive) but varies in magnitude and significance.
- **RQ3.** GDI gives a usable within-panel ranking at adequate n, but not one that is stable across scales.
- **RQ4.** The disparity splits into a dominant geographic component and a weak name component that do not simply add.

Bottom line

Geographic-axis bias in clinical LLMs is real but model-heterogeneous. Detecting it reliably needs real clinical text and a sample size justified by a power analysis.

Limitations and future work

Limitations

- Single seed; no multi-seed variance estimate yet.
- Pilot gold labels are team-assigned, not clinician-validated.
- Findings scoped to OncQA oncology text and three South regions.

Future work

- Scale to r/AskaDocs and USMLE+Derm.
- Exercise the Southeast Asia and MENA regions.
- Multi-seed runs; intersectional (gender $\times$ region) slices.
- Board-certified clinician inter-rater agreement.

Thank you

Questions and discussion

Every result in this talk traces to a timestamped run directory.
Full methods, tables, and reproducibility notes are in the final report.